Code No: 135AE **R16**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech III Year I Semester Examinations, February - 2022
DATA COMMUNICATION AND COMPUTER NETWORKS
(Common to CSE, IT)

Time: 3 hours **Max. Marks: 75**

Answer any five questions
All questions carry equal marks
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1.a) List and explain about various layers in TCP/IP model.
b) Discuss about Light wave transmission along with its limitations. [7+8]

2.a) Give the Motivation for Layered Architecture Design of a Computer Network.
b) Categorize the Multiplexing techniques. Explain any technique designed for digital signals. [7+8]

3.a) Explain about various multiple access protocols.
b) Compare and contrast the Go-Back-N ARQ Protocol with Selective-Repeat ARQ. [7+8]

4.a) Apply the CRC method to detect an error/s if any, with the given data word "1010011110", divisior "10111" and received codeword "10100111101100". Give its correcting method.
b) What is the purpose of CSMA/CD? Explain it. [7+8]

5.a) If a router has 20 entries in its group table, should it send 20 different queries periodically or just one? Justify your answer.
b) List the transition strategies to move from IPv4 to IPv6. When is each strategy used? [8+7]

6.a) Distinguish between Distance Vector Routing and Link State Routing Protocols.
b) What is the format of IPv4 header? Describe the significance of each field. [8+7]

7.a) Discuss about the TCP sliding window algorithm for Error control.
b) Categorize the Congestion control techniques. Explain in brief about any one category. [7+8]

8.a) Explain how FTP can be used to solve the problems in transferring the file from one system to another system.
b) Define Network Management. List and explain the functions of Network Management. [8+7]

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